

Atho Monetary Policy and 150-Year Supply Schedule

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The Platinum Standard of the Quantum Age

1. Executive Summary

This attachment summarizes the monetary behavior enforced by the current Atho codebase in this repository. The active implementation uses:

- a **100-second** target block time
- **12 decimals**
- **1 ATHO = 1,000,000,000,000 atoms**
- an initial block reward of **5 ATHO**
- a halving interval of **1,260,000 blocks**
- a perpetual tail reward of **0.625 ATHO**
- **no finite maximum ATHO supply cap** in current consensus

Full nodes enforce block subsidy correctness, fee accounting, ownership validation, and spendability rules independently. The schedule below summarizes subsidy issuance over 150 years under the code as written today.

2. Network Monetary Parameters

Parameter	Current Code Value
Target block time	100 seconds

Parameter	Current Code Value
Blocks per hour	36
Blocks per day	864
Blocks per year	315,360
Decimals	12
Atoms per ATHO	1,000,000,000,000
Initial block reward	5 ATHO
Halving interval	1,260,000 blocks
Tail reward	0.625 ATHO
Tail emission start	Block 3,780,000
Coinbase maturity	100 blocks
Standard confirmations	6 across all networks
Proof-of-Work hash	SHA3-384
Signature scheme	Falcon-512
Emission type	Halving schedule with perpetual tail emission
Supply enforcement rule	Per-block subsidy validation by full nodes
Fixed max supply cap	None in the current code

3. Emission Formula

The live repository implements the following reward rule:

$$\text{reward}(\text{height}) = \max(5 / 2^{\text{floor}(\text{height} / 1,260,000)}, 0.625) \text{ ATHO}$$

In plain language:

1. Subsidy begins at 5 ATHO.
2. It halves every 1,260,000 blocks.
3. Once the halved value would drop below 0.625 ATHO, the reward floors at 0.625 ATHO permanently.
4. No finite cap currently stops emission after tail activation.

4. 150-Year Supply Schedule

Table 1

Atho 150-Year Monetary Supply Projection Under the Current Emission Policy

Year	Estimated Block Height	Block Reward (ATHO)	Annual Emission (ATHO)	Cumulative Supply (ATHO)	% of Enforced Max Supply	Remaining to Enforced Max Supply
1	315,360	5	1,576,800	1,576,800	N/A	N/A
2	630,720	5	1,576,800	3,153,600	N/A	N/A
3	946,080	5	1,576,800	4,730,400	N/A	N/A
4	1,261,440	2.5	1,573,200	6,303,600	N/A	N/A
5	1,576,800	2.5	788,400	7,092,000	N/A	N/A
10	3,153,600	1.25	394,200	10,242,000	N/A	N/A

Year	Estimated Block Height	Block Reward (ATHO)	Annual Emission (ATHO)	Cumulative Supply (ATHO)	% of Enforced Max Supply	Remaining to Enforced Max Supply
15	4,730,400	0.625	197,100	11,619,000	N/A	N/A
20	6,307,200	0.625	197,100	12,604,500	N/A	N/A
25	7,884,000	0.625	197,100	13,590,000	N/A	N/A
30	9,460,800	0.625	197,100	14,575,500	N/A	N/A
40	12,614,400	0.625	197,100	16,546,500	N/A	N/A
50	15,768,000	0.625	197,100	18,517,500	N/A	N/A
75	23,652,000	0.625	197,100	23,445,000	N/A	N/A
100	31,536,000	0.625	197,100	28,372,500	N/A	N/A
125	39,420,000	0.625	197,100	33,300,000	N/A	N/A
150	47,304,000	0.625	197,100	38,227,500	N/A	N/A

Note. Projection assumes the current code constants remain unchanged for the entire horizon. It uses 100-second blocks, 315,360 blocks per year, and subsidy issuance only. Because the current code does not enforce a finite maximum ATHO supply, cap-based percentage and remaining-supply fields are not applicable.

5. Notes and Assumptions

1. The projection is based on the active code, not on older planning text.
2. Yearly checkpoints represent end-of-year heights under the fixed target cadence.

3. Block timestamps in practice may vary, but the subsidy function is height-based.
4. Transaction fees are excluded because they do not mint new ATHO.
5. Tail emission begins at block 3,780,000, or roughly year 11.99 under current constants.
6. The current repository does not enforce a 78,000,000 ATHO maximum supply.

6. Final Policy Statement

Atho's monetary policy is designed to be deterministic, auditable, and enforceable by every full node. Under the current code, no block may create ATHO beyond the consensus-defined subsidy schedule for its height, but the present implementation does not yet enforce a finite maximum ATHO supply cap.

7. Founder-Review Mismatch Note

The current code now matches the 100-second block cadence, 6-confirmation policy, 100-block coinbase maturity, and revised $5 \rightarrow 2.5 \rightarrow 1.25 \rightarrow 0.625$ subsidy path described in updated monetary planning materials. Any public statement claiming a fixed 78,000,000 ATHO cap would still be inaccurate unless the code changes to enforce one.